PSEUDO-COHERENT COMPLEXES WITH BOUNDED TOR AMPLITUDE OVER A FIBREWISE FINITE-GLOBAL-DIMENSION BASE ARE PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 20174-20188 (canonical tag 068X) and the counted proof is the proof environment at source lines 20190-20221, which immediately follows it and is its own proof.

Editorial note. The proof refers twice inside the chapter: to lemma-finite-gl-dim-tor-dimension at source line 17287 for the fibrewise tor amplitude estimate and to lemma-check-perfect-pointwise at source line 20032 for the fibrewise criterion of perfectness. Both are navigation pointers and their roles are recorded in the external reference reasons; the displayed computation of the fibres over primes of B is inside the counted span.

Editorial note. The lead-in at source lines 20170-20172, announcing the case of a polynomial ring extension, is outside the counted spans, as is the section text that follows the proof.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- A complex over a ring of global dimension at most d whose cohomology is concentrated in degrees a to b has tor amplitude in the shifted range a - d to b. invoked at proof lines 20201-20204, where the fibre K tensor_A kappa(p) over the residue field is estimated using the finite global dimension hypothesis. Converts the fibrewise cohomology bound into the fibrewise tor amplitude bound that has to be propagated to the fibres over primes of B.
- Perfectness of a pseudo-coherent complex over a ring can be checked on the fibres at the primes, by testing the vanishing of the derived tensor products with the residue fields. invoked at proof lines 20217-20220, where the vanishing of the shifted cohomology for all primes q of B gives the

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conclusion. Turns the pointwise computation just performed into the global perfectness statement with the stated tor amplitude range.
- For a flat ring map A to B and a complex of B-modules that is flat over A, the derived tensor product with a B-module agrees with the ordinary tensor product, and localization at a prime of B is computed through the residue field of the prime below. used at proof lines 20191-20205 and 20206-20216. Justifies replacing the derived tensors over A and over B by ordinary ones and the displayed chain of identifications for K tensor^L_B kappa(q).

Source reference. Original lemma 066P: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17287–17287. The reference is used at proof lines 20201-20204 to turn the fibrewise cohomology bound into the fibrewise tor amplitude bound over A; the rest of the proof, namely the reduction to finite free complexes and the computation of the fibres over B, is inside the counted span.

Source reference. Original lemma 068V: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20032–20032. The reference is the fibrewise criterion invoked at proof lines 20217-20220 to conclude perfectness; the vanishing of the cohomology of every fibre, which is the condition the criterion tests, is established inside the counted span at lines 20205-20216.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $a, b \in \mathbf{Z}$. Let $d \geq 0$. Let $K^\bullet$ be a complex of B-modules. Assume*

(1) *the ring map $A \to B$ is flat,*
(2) *for every prime $\mathfrak{p} \subset A$ the ring $B \otimes_A \kappa(\mathfrak{p})$ has finite global dimension $\leq d$,*
(3) *$K^\bullet$ is pseudo-coherent as a complex of B-modules, and*
(4) *$K^\bullet$ has tor amplitude in $[a, b]$ as a complex of A-modules.*

Then $K^\bullet$ is perfect as a complex of B-modules with tor amplitude in $[a - d, b]$.

Proof. We may assume that $K^\bullet$ is a bounded above complex of finite free B-modules. In particular, $K^\bullet$ is flat as a complex of A-modules and $K^\bullet \otimes_A M = K^\bullet \otimes^{\mathbf{L}}_A M$ for any A-module M. For every prime $\mathfrak{p}$ of A the complex

$$K^\bullet \otimes_A \kappa(\mathfrak{p})$$

is a bounded above complex of finite free modules over $B \otimes_A \kappa(\mathfrak{p})$ with vanishing H^i except for $i \in [a, b]$. As $B \otimes_A \kappa(\mathfrak{p})$ has global dimension d we see from Lemma 066P that $K^\bullet \otimes_A \kappa(\mathfrak{p})$ has tor amplitude in $[a - d, b]$. Let $\mathfrak{q}$ be a prime of B lying over $\mathfrak{p}$. Since $K^\bullet \otimes_A \kappa(\mathfrak{p})$ is a bounded above complex of free $B \otimes_A \kappa(\mathfrak{p})$-modules we see that

$$\begin{aligned} K^\bullet \otimes^{\mathbf{L}}_B \kappa(\mathfrak{q}) &= K^\bullet \otimes_B \kappa(\mathfrak{q}) \\ &= (K^\bullet \otimes_A \kappa(\mathfrak{p})) \otimes_{B \otimes_A \kappa(\mathfrak{p})} \kappa(\mathfrak{q}) \\ &= (K^\bullet \otimes_A \kappa(\mathfrak{p})) \otimes^{\mathbf{L}}_{B \otimes_A \kappa(\mathfrak{p})} \kappa(\mathfrak{q}) \end{aligned}$$

Hence the arguments above imply that $H^i(K^\bullet \otimes^{\mathbf{L}}_B \kappa(\mathfrak{q})) = 0$ for $i \notin [a - d, b]$. We conclude by Lemma 068V. □